Genese: execution architecture and measured performance

Brain handles decisions. World/ECS executes effects. Partitions overlap preparation, inference and ledger updates.

Native execution capabilities

Paper Table S13 · 10 mechanisms × 6 frameworks

Capability	Genese	OneSim	OASIS	MiroFish	AgentSociety	MatrAIx
Array-based ECS storage	**Yes**	N/D	N/D	N/D	N/D	N/D
Synchronized state persistence	**Yes**	Yes	No	No	No	No
Native action-rule enforcement	**Yes**	Partial	Yes	Yes	Yes	Partial
Policy-intent / world-effect separation	**Yes**	No	No	No	No	No
Atomic state–event commit	**Yes**	No	No	No	No	No
World-step isolation / rollback	**Yes**	No	No	No	Partial	Partial
Recovery / resume	**Yes**	No	No	No	Yes	Yes
Execution / connection pooling	**Yes**	Yes	No	No	Yes	No
Task admission / flow control	**Yes**	Partial	Yes	Yes	Yes	Yes
Decision / trial concurrency control	**Yes**	Partial	Yes	Yes	Yes	Yes

Yes: implemented Partial: part of the defined mechanism No: absent in the inspected path N/D: not documented

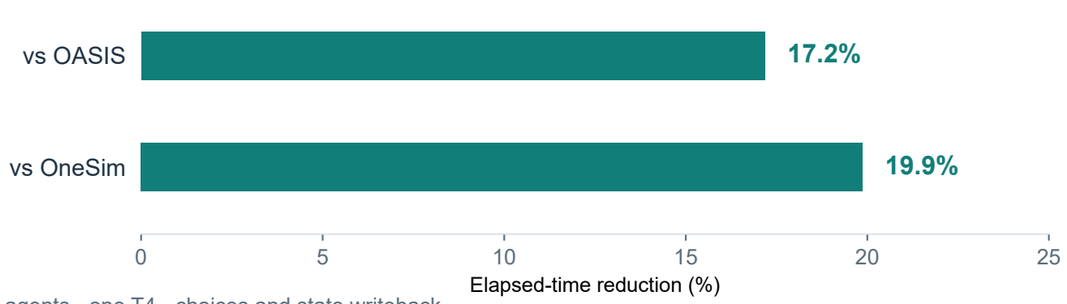
Action rules: OASIS/MiroFish platform rules; AgentSociety permissions and workspace. Resume: AgentSociety checkpoints; MatrAIx job progress.

OneSim: PostgreSQL supports synchronized persistence; event/interface and decision-flow controls are Partial. MatrAIx: questionnaire-output checks are Partial.

AgentSociety and MatrAIx isolate independent tasks; whole-world rollback is a separate mechanism.

17–20% less execution time

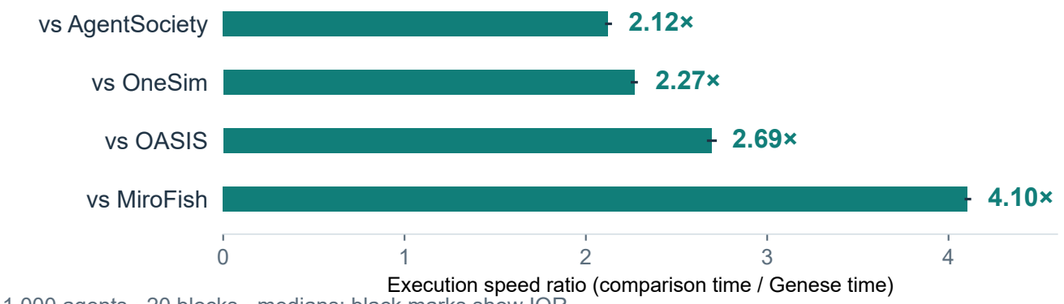
Actual sharp-421m decisions · shared model allocation



1,000 agents · one T4 · choices and state writeback

2.12–4.10× execution speed

Fixed immediate decision backend · relative to each engine



1,000 agents · 20 blocks · medians; black marks show IQR

MatrAIx parallel questionnaire adapter: 1,062.153 s; Genese parallel partitions: 2.743 s on the immediate-backend workload.